

Additional file 1

A metadata-matched binned baseline halves the apparent peak-level advantage in MS/MS substructure prediction

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This file reports the per-arm and per-seed results behind every figure in the main text, the full bootstrap distributions, and the peak-absorption analysis broken down by acquisition condition. Every table here is also provided as a machine-readable CSV in the repository under `results/`, and every number is regenerated from saved run artefacts by `scripts/make_supplementary.py` and `scripts/make_si_figures.py`. Section S8 states both controls in full, so that the method can be applied without the repository.

S1 Per-arm, per-seed metrics

Table S1 gives every trained arm on both corpora. Twenty-seven training runs: five arms $\times$ three seeds on the open corpus, four $\times$ three on the single-library corpus. Parameter counts are identical between M1R and M1 by construction, and the binned baseline M0 carries 65% more parameters than either.

Every arm in both ladders is conditioned on the acquisition covariates (collision energy, adduct, polarity, instrument). In earlier versions of this work M0 alone was not: its encoder accepted the covariate vector and discarded it, so the binned baseline was the only arm predicting without knowing how the spectrum had been acquired. All six M0 rows above are refits under the corrected baseline and supersede the figures reported previously; the main text states what the correction cost the headline claim.

Conditioning was worth +0.0288 to M0 on the open corpus and +0.0142 on the single-library corpus. That the gain roughly doubles with the corpus’s acquisition diversity — 66 instrument types and 112 adducts against two and sixteen — is what one would expect if the covariates are doing the work we ascribe to them, and we note it as a consistency check rather than a result.

S2 Per-seed effect sizes and ordering checks

Labels. The tables below use the short names *interface*, *attention* and *encoder* for M1D – M0, M1R – M1D and M1R – M0, which are the keys under which the analysis scripts emit them. The main text calls the same three contrasts the *binned token-encoder package*, the *attention-bearing versus attention-free comparison* and the *rounded token-encoder package*, because none of the three changes only the component its short name suggests: the arms either side of each also differ in pooling, in feed-forward width, or in how the acquisition covariates are injected. The main text’s names are the accurate ones. The short names are retained here so that the tables match the column keys in `results/`.

Each effect is the difference between two arms trained under the *same* seed, so the three values per row are matched rather than independent samples (Table S2). Where the main text claims an ordering between two effects, it is checked within each seed; Table S3 reports those checks.

Table S1: Held-out metrics for every arm and seed. Macro AUPRC is the primary metric; chance is 0.0085 on the open corpus and 0.0093 on the single-library corpus. Machine-readable: `results/si_table_s1_arm_metrics.csv`.

corpus	arm	seed	parameters	best epoch	macro AUPRC	macro AUROC	Brier
open	M0	2026	1,172,752	13	0.1433	0.8815	0.00531
open	M0	7	1,172,752	14	0.1416	0.8793	0.00534
open	M0	13	1,172,752	14	0.1406	0.8788	0.00535
open	M1D	2026	714,528	12	0.1616	0.8930	0.00522
open	M1D	7	714,528	13	0.1569	0.8915	0.00527
open	M1D	13	714,528	14	0.1585	0.8940	0.00524
open	M1R	2026	712,336	14	0.1591	0.8991	0.00525
open	M1R	7	712,336	13	0.1570	0.8980	0.00525
open	M1R	13	712,336	14	0.1541	0.8960	0.00526
open	M1	2026	712,336	14	0.1731	0.9063	0.00513
open	M1	7	712,336	13	0.1709	0.9063	0.00517
open	M1	13	712,336	13	0.1687	0.9068	0.00514
open	M2	2026	829,584	14	0.1899	0.9072	0.00512
open	M2	7	829,584	14	0.1893	0.9052	0.00511
open	M2	13	829,584	12	0.1871	0.9090	0.00512
single library	M0	2026	1,172,752	14	0.0631	0.8096	0.00689
single library	M0	7	1,172,752	14	0.0621	0.8105	0.00688
single library	M0	13	1,172,752	13	0.0608	0.8109	0.00688
single library	M1R	2026	712,336	13	0.0867	0.8457	0.00673
single library	M1R	7	712,336	14	0.0856	0.8474	0.00672
single library	M1R	13	712,336	14	0.0840	0.8470	0.00675
single library	M1	2026	712,336	14	0.1119	0.8734	0.00654
single library	M1	7	712,336	13	0.1085	0.8717	0.00656
single library	M1	13	712,336	13	0.1162	0.8759	0.00652
single library	M2	2026	829,584	13	0.1460	0.8861	0.00640
single library	M2	7	829,584	13	0.1428	0.8824	0.00644
single library	M2	13	829,584	14	0.1444	0.8837	0.00641

Table S2: Effect sizes per seed, as differences in held-out macro AUPRC.

corpus	effect	seed 2026	seed 7	seed 13	mean	SD
open	interface (M1D – M0)	+0.0183	+0.0153	+0.0179	+0.0172	0.0016
open	attention (M1R – M1D)	−0.0025	+0.0001	−0.0045	−0.0023	0.0023
open	encoder (M1R – M0)	+0.0159	+0.0154	+0.0134	+0.0149	0.0013
open	mass axis (M1 – M1R)	+0.0140	+0.0139	+0.0147	+0.0142	0.0004
open	aggregation (M2 – M1)	+0.0168	+0.0184	+0.0184	+0.0179	0.0009
single library	encoder	+0.0236	+0.0235	+0.0232	+0.0234	0.0002
single library	mass axis	+0.0252	+0.0229	+0.0322	+0.0268	0.0049
single library	aggregation	+0.0341	+0.0343	+0.0282	+0.0322	0.0035

Table S3: Ordering checks. Each entry is the within-seed difference between two effects; an ordering is reported in the main text only where it holds in all three seeds.

corpus	comparison	seed 2026	seed 7	seed 13	mean	verdict
open	encoder – mass axis	+0.0019	+0.0015	−0.0012	+0.0007	reverses
open	encoder – aggregation	−0.0009	−0.0030	−0.0049	−0.0029	3/3 consistent
open	mass axis – aggregation	−0.0028	−0.0045	−0.0037	−0.0037	3/3 consistent
single library	encoder – mass axis	−0.0016	+0.0006	−0.0090	−0.0033	reverses
single library	encoder – aggregation	−0.0105	−0.0108	−0.0049	−0.0087	3/3 consistent
single library	mass axis – aggregation	−0.0089	−0.0114	+0.0041	−0.0054	reverses

Three things are worth reading off this table. The encoder-against-mass-axis row reverses on both corpora independently, which is why the main text declines to order those two anywhere. The encoder-against-aggregation row is consistent on both and negative on both, which is why aggregation is the one ranking the paper does claim. And the final row is why the main text ranks the mass axis against aggregation on the open corpus and declines to on the single-library corpus.

S3 Bootstrap distributions

Figure S1 shows the full distribution behind every interval reported in the main text: 5,000 resamples of the mean paired per-target difference, over targets, after averaging each target’s average precision across the three seeds. Table S4 summarises them.

Table S4: Bootstrap summaries. Machine-readable draws: `results/si_table.s3_bootstrap.draws.csv.gz`.

corpus	effect	mean	SD	2.5%	97.5%
open	interface	0.01727	0.00360	0.01020	0.02419
open	attention	−0.00230	0.00221	−0.00665	0.00196
open	encoder	0.01498	0.00362	0.00798	0.02210
open	mass axis	0.01415	0.00230	0.00969	0.01865
open	aggregation	0.01785	0.00222	0.01360	0.02233
single library	encoder	0.02344	0.00194	0.01972	0.02729
single library	mass axis	0.02678	0.00243	0.02208	0.03168
single library	aggregation	0.03219	0.00243	0.02751	0.03696

What these intervals do and do not cover

Each interval resamples targets, having first averaged average precision over the three seeds. It therefore describes uncertainty from the finite set of targets, not from initialisation. Seed variability is reported separately as the SD column of Table S2, and the two should be read together: on the single-library corpus the mass-axis effect has a seed SD of 0.0049, comparable to the width of its bootstrap interval, which is the statistical form of the ordering failure in Table S3. A variance-components treatment with targets and seeds as crossed random effects would fold both into one interval and we regard it as the right analysis for a larger replicate count.

The attention arm

M1D removes self-attention from every block while holding the mass axis, the peak tokens and the acquisition conditioning fixed. Two further things change with it. A learned pooling token

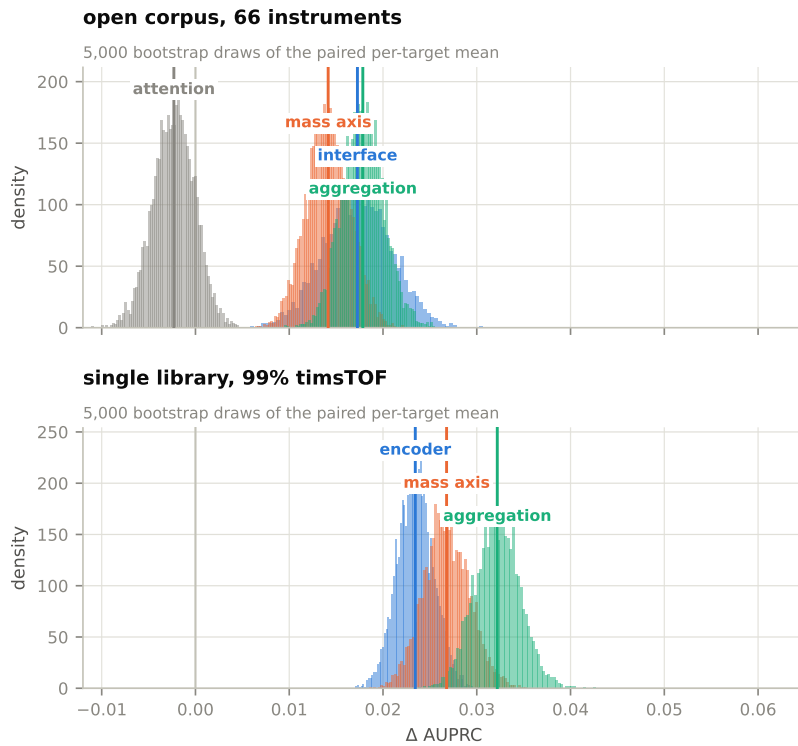


Figure S1: Bootstrap distributions on each corpus. The open panel shows the M1R – M0 contrast split into its two parts: the M1D – M0 distribution sits far from the others while the attention distribution is centred on zero. On the single-library corpus, which has no attention arm, the composite encoder effect is shown instead. The encoder and mass-axis distributions overlap heavily there, and the mass-axis and aggregation ones likewise, which is the same fact Table S3 records as two sign reversals; only the encoder and aggregation distributions separate.

cannot aggregate position-wise tokens once attention is gone, so M1D pools by a masked mean; and removing attention frees 66,048 weights per block, so the feed-forward is widened to 548 units in this arm alone to keep the arm from also being the smaller model (714,528 parameters against 712,336). M1R – M1D therefore compares two configurations rather than pricing attention on its own, and the main text reports it as such. An exact ablation holding pooling and feed-forward width fixed on both sides is the experiment that would price attention, and it has not been run. What can be said from these runs is that this contrast is the only one in the study whose interval includes zero, and the only one positive in fewer than three seeds of three.

S4 Per-target average precision

`results/si_table_s2_per_target_ap.csv` gives average precision for every scored target, for every arm, both as the seed-averaged value used in the paired analysis and per seed: 433 targets on the open corpus and 477 on the single-library corpus, 910 rows in total. All paired differences in the main text are recoverable from this file by subtraction.

S5 Absorption by acquisition condition

The main text reports that 23.6% of the open corpus’s held-out peaks fall into a 0.5 Da bin already occupied. Figure S2 breaks that down by adduct, collision energy and instrument family.

Absorption at 0.5 Da by acquisition stratum, open corpus held-out spectra

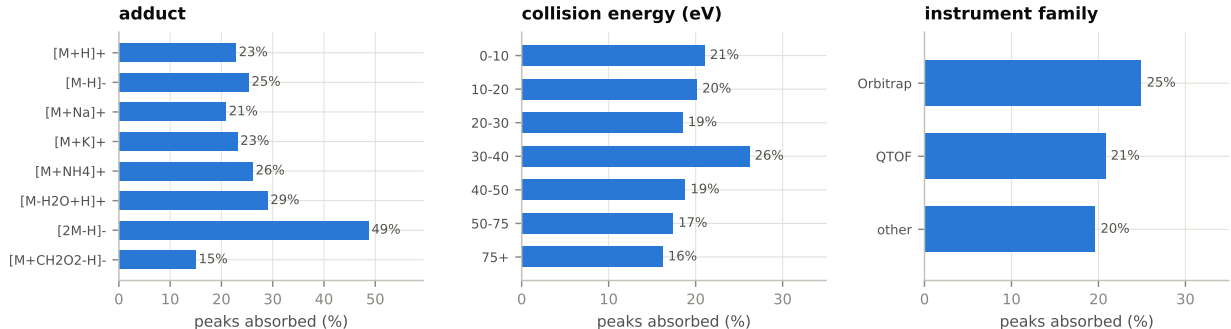


Figure S2: Absorption at 0.5 Da by acquisition stratum, over the open corpus’s held-out spectra. Strata with fewer than 20,000 peaks are omitted.

Absorption tracks peak density rather than any chemical property of the adduct. The highest rate among common adducts is $[2M-H]^-$ at 48.7%, whose spectra are dimeric and correspondingly crowded; $[M+H]^+$, which supplies 62% of held-out spectra, sits at 22.8%. Across collision energies the range is 16% to 26% with no monotone trend. Orbitrap spectra lose 24.9% against QTOF’s 20.8%, again following density (52.6 against 47.2 peaks per spectrum) rather than resolving power.

This is worth stating because it constrains the interpretation in the main text. Binning costs more where peaks are dense, and density varies with acquisition conditions that have nothing to do with how much mass precision the instrument actually delivered.

Machine-readable, all under `results/`:

`si_absorption_by_adduct.csv`
`si_absorption_by_collision_energy.csv`
`si_absorption_by_instrument_family.csv`

S6 Bin-width sweep

Table S5 gives the sweep in full. Every arm uses the M1 encoder at an identical parameter count; only the width of the mass grid changes. One run per width, on the single-library corpus.

Table S5: Bin-width sweep, single seed, single-library corpus.

bin width (Da)	peaks absorbed	macro AUPRC	share of the 0.5 Da gap closed
0.5	37.5%	0.0867	0%
0.25	36.8%	0.0818	−19%
0.1	33.1%	0.0801	−26%
0.05	25.9%	0.0909	17%
0.01	7.5%	0.1030	65%
full precision	0.0%	0.1119	100%

The absorption column here is computed over the competition test spectra, which are denser than the corpus’s own held-out spectra (72.5 against 58 peaks per spectrum); the corresponding held-out figures are 34.4% at 0.5 Da falling to 6.5% at 0.01 Da, and those are the numbers the main text uses.

S7 Reproduction

Software: Python 3.14, PyTorch 2.11.0+cu128, RDKit 2026.3.6, NumPy 2.4.4, pandas 2.3.3, pyarrow 24.0.0, on a single NVIDIA RTX 3070. Exact pins for every dependency are in `requirements-lock.txt`.

The full path from the corpus file to every number in this file is

```

python scripts/make_open_subset.py --root .
python -m dbf2 prepare --root . --dataset open
python -m dbf2 ablate --root . --dataset open \
    --rungs M0 M1D M1R M1 M2 --epochs 15
python -m dbf2 ablate --root . --dataset open \
    --rungs M0 M1D M1R M1 M2 --epochs 15 --seed 7
python -m dbf2 ablate --root . --dataset open \
    --rungs M0 M1D M1R M1 M2 --epochs 15 --seed 13
python scripts/make_supplementary.py
python scripts/make_si_figures.py

```

with the same three commands minus `--dataset open` and minus M1D for the single-library corpus, which carries four arms, and

```

for w in 0.25 0.1 0.05 0.01; do
    python -m dbf2 train --root . --rung M1 \
        --quantise $w --tag W$w --epochs 15
done

```

for the sweep. `python -m pytest src/dbf2/tests` runs 71 unit tests, of which 14 cover the rounding control — including that M1R and M1 have identical parameter counts and that their configurations differ in exactly one field — and five check that every arm responds to its acquisition covariates, the regression described in S1. `pytest packages/msms-controls` runs a further 23 against the released library. Neither suite requires the corpus.

A note on what the subset builder reads

`make_open_subset.py` selects the GNPS, MassBank and MoNA rows out of the aggregated CASMI 2026 file. Those spectra are openly licensed at their sources, but reproducing this work as written still requires that aggregated file. Fetching the records from GNPS, MassBank and MoNA directly is recorded as an open item in the repository. `docs/corpus_composition.md` reports each source library’s contribution and states which parts are redistributable.

S8 The two controls in full

Both controls are released as `msms-controls`, a NumPy-only library under the MIT licence. They are short enough to state here as well, so that the method does not depend on a repository still resolving. Neither is specific to this model, this corpus or this deep-learning framework.

The rounding control

Given one spectrum’s peaks (m_k, I_k) , its precursor m^{prec} , and the grid width w used by the binned baseline:

```
# 1. assign peaks to the baseline’s cells, and move each to its centre
c_k      = floor(m_k / w)
m_out    = (unique(c_k) + 0.5) * w

# 2. merge the peaks that now share a cell, using the accumulator the
#     baseline uses. For a sqrt-intensity binned vector that is
I_out[c] = ( sum of sqrt(I_k) over peaks k with c_k = c ) ** 2

# 3. snap the precursor to the SAME grid
m_prec_out = (floor(m_prec / w) + 0.5) * w
```

The resulting peaks pass through the unmodified peak-set encoder. No peak is added or removed; only the mass axis is degraded and colliding peaks are combined.

Step 3 is the one that is easy to omit and costly to omit. Most peak encoders derive a neutral loss as $m^{\text{prec}} - m_k$. If the peaks are quantised while the precursor keeps full precision, that subtraction returns the precursor’s mass defect to every token, and the arm intended to carry only binned mass information silently regains a full-precision channel. The control then understates the value of the mass axis, and nothing in the training curves reveals it. A unit test asserts that two precursors falling in one cell give identical embeddings under the control and different embeddings without it.

Step 2 must match the baseline, not merely be reasonable. Accumulating raw intensity where the baseline accumulates its square root changes the intensity scale as well as the mass axis, which reintroduces exactly the kind of two-factor comparison the control exists to remove.

The conditioning-parity check

Given the arms of the ablation, one fixed spectrum, and two settings c_1, c_2 of the acquisition covariates:

```
for each arm a:
    y1 = predict(a, spectrum, covariates = c1)
    y2 = predict(a, spectrum, covariates = c2)
    response[a] = max |y1 - y2|

fail, naming every offender, if any response[a] <= tol
```

Choose c_1 and c_2 to differ substantially — a low against a high collision energy, two adducts, two instruments — since an arm may legitimately be nearly invariant to a small change. An arm whose response is zero is not a weaker model but a different experiment: every contrast measured against it also prices the information it could not see. That is the failure reported in S1, and it cost half of what a two-arm comparison here would have attributed to the spectral representation.

One caveat makes the difference between a real check and a vacuous one. FiLM and adaptive layer normalisation are commonly initialised so that the modulation is the identity at step 0, which makes an untrained model return exactly zero for every arm whether or not the conditioning is wired correctly. Perturb the conditioning parameters off their initialisation before testing, or test a trained checkpoint.